TP6 Hypothesis Testing — Group Assignment

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Exercise 1

Question 1

The null hypothesis is $H_0 : \mu \geq 8\%$ and the alternative hypothesis is $H_1 : \mu < 8\%$

Question 2

The type of test that should be used here is a one-sided test (less than). This is because we only care about testing in one direction, not about testing if there is a difference

Exercise 2

Question 1

The null hypothesis for testing whether the training has an effect on stress level is $H_0 : \mu_{\text{before}} = \mu_{\text{after}}$ and the alternative hypothesis is $H_1 : \mu_{\text{before}} \neq \mu_{\text{after}}$.

Question 2

"Rejecting H_0" in this example implies that the mindfulness training program has an effect on the average stress level. However it does not indicate if the stress level increases or decreases.

Exercise 3: Hypothesis Test for a New Teaching Method

Setup: let $\mu_{diff} = \mu_{new} - \mu_{traditional}$ be the true mean difference in exam scores between the new and the traditional teaching method. In the sample, the new method has mean 78 and the traditional method has mean 75 (observed difference 3), and the test gave $p = 0.12$ at significance level $\alpha = 0.05$.

1. Null and alternative hypotheses

$$\mathbf{H_0}: \mu_{diff} = 0 \quad \text{(the two methods have the same true mean score)}$$

$$\mathbf{H_1}: \mu_{diff} \neq 0 \quad \text{(the true mean scores differ)}$$

The question asks whether the new method leads to a *different* average score, so H_1 does not specify a direction.

2. Type of test

A two-sided test should be used. Reasons:

• The research question is whether the average score is *different*, not whether it is higher or lower. A two-sided alternative ($\mu_{diff} \neq 0$) matches this wording exactly.

• A new method could plausibly help *or* hurt. A two-sided test detects a deviation from H_0 in either direction, whereas a one-sided test would ignore effects in the other direction.

• The direction of the test must be fixed from the research question *before* looking at the data. Choosing a one-sided test just because the sample difference happens to be positive ($78 > 75$) would bias the test in favor of rejecting H_0.

Conclusion of the test

Since $p = 0.12 > \alpha = 0.05$, we **fail to reject $\mathbf{H_0}$**. There is not enough evidence at the 5% level that the new method changes the mean exam score. The observed 3-point difference is consistent with random sampling variation. (This does not prove the methods are equal; it only means the data do not provide sufficient evidence of a difference.)

Exercise 4

1. Hypotheses

- **Null hypothesis (H_0):** $\mu_B = \mu_A$ (or $\mu_B \leq \mu_A$, meaning Model B does not improve true prediction accuracy compared to Model A).
- **Alternative hypothesis (H_1):** $\mu_B > \mu_A$ (meaning Model B has a higher true prediction accuracy than Model A).

2. Decision at $\alpha = 0.05$

Yes, the data scientist should claim that Model B is significantly better than Model A.

Explanation: Since the p-value (0.04) is less than the significance level $\alpha = 0.05$ ($0.04 < 0.05$), we reject the null hypothesis H_0 in favor of the alternative hypothesis H_1. This indicates that observing an improvement at least this large by random chance alone is unlikely under H_0, providing statistically significant evidence at the 5% level that Model B improves accuracy.

3. Decision at $\alpha = 0.01$

If the significance level were set to $\alpha = 0.01$, the decision would change: we would **fail to reject** H_0.

Explanation: Because the p-value (0.04) is greater than the stricter significance level $\alpha = 0.01$ ($0.04 > 0.01$), the test result is no longer statistically significant at the 1% level. There would not be sufficient statistical evidence at the 1% significance level to claim that Model B performs better than Model A.

Exercise 5

Problem 1

Problematic statement: "Because the observed average loading time was lower after optimization, we selected a one-sided test."
Issue: The choice between a one-sided and a two-sided test must be made *before* looking at the sample data, based on the research question or prior claim. Selecting a one-sided test after observing the direction of the sample data (post-hoc) invalidates the test's Type I error rate.
Correction: "Because our goal is to test the web engineer's prior claim that the optimization reduces average page-loading time, we specified a one-sided test before collecting and examining the data."

Problem 2

Problematic statement: "The p-value of 0.07 means that there is a 7% probability that the optimization has no effect."
Issue: The p-value is not the probability that the null hypothesis (H_0) is true. Instead, it is the probability of observing a sample result at least as extreme as the one obtained, *assuming* that the null hypothesis (H_0) is true.
Correction: "The p-value of 0.07 means that if the optimization truly had no effect on average

page-loading time (H_0 were true), there would be a 7% probability of observing a sample reduction in average loading time at least as large as the one we observed."

Problem 3

Problematic statement: "Because $p > 0.05$, we accept H_0 and conclude that the optimization does not improve loading performance."
Issue: In hypothesis testing, we never "accept H_0" or conclude that H_0 is proven true. A p-value greater than α only means that we lack sufficient evidence to reject H_0.
Correction: "Because $p = 0.07 > 0.05$, we fail to reject H_0 at the 5% significance level. There is insufficient statistical evidence at the 5% level to conclude that the optimization reduces average page-loading time."

Problem 4

Problematic statement: "Therefore, it should not be deployed."
Issue: A decision should not be based solely on whether a p-value crosses an arbitrary $\alpha = 0.05$ cutoff (especially with a borderline p-value of 0.07). It ignores practical significance (effect size and confidence interval), the statistical power of the test, and the cost-benefit trade-off of deployment.
Correction: "While the reduction is not statistically significant at the 5% level ($p = 0.07$), the decision to deploy should also consider the estimated effect size (confidence interval), the statistical power of the study, and the engineering cost versus potential performance benefit."